



How to Build a Vector RAG?

1. Ingest (Collect) & Preprocess Data



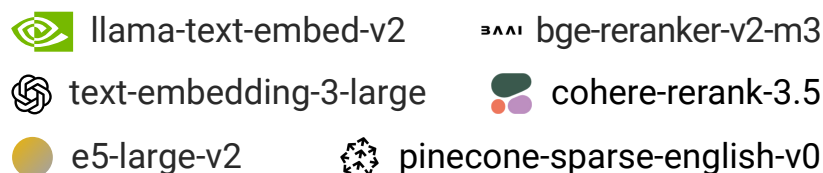
2. Split Into Chunks



Best Practice

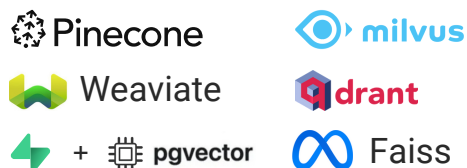
Use semantic-aware splitting first, then apply token-aware constraints. That's how you preserve meaning and avoid token overflow.

3. Generate Embeddings



4. Store in Vector DB & Index

Vector DBs



Document DBs



Knowledge Graphs



5. Retrieve

Vector (dense, semantic) and/or docs (sparse) retrieval:



Helpful algorithms

- BM25 (Best Matching 25) - for vector match
- Reciprocal Rank Fusion (RRF) - to combine multiple ranked lists

6. Orchestrate the Pipeline



7. Select LLMs for Generation



LLM Gateways



8. Add Observability

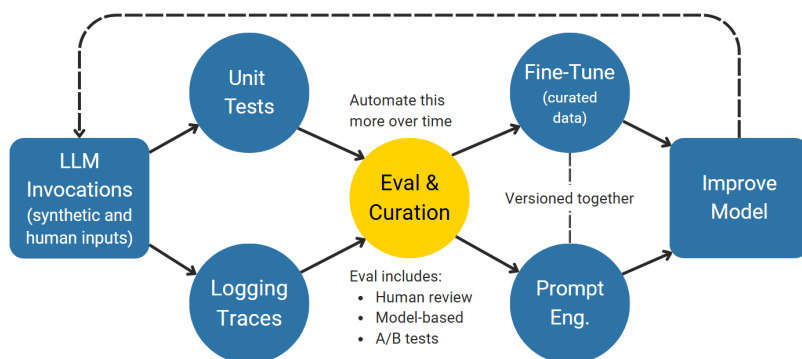


9. Evaluate Retrieval & Generation

Standard retrieval metrics:

- Precision(K)
- Recall(K)

Evaluating generation (LLMs):



3	A/B testing	cost	★★★	frequency	★★★
2	Model & Human Eval	cost	★★★	frequency	★★★
1	Unit Tests	cost	★★★	frequency	★★★

START HERE

S1: Log Traces
S2: Look at Traces
S3: Model & Human

Step 1: Write Unit Tests
Step 2: Create Test Cases
Step 3: Run & Track Tests Regularly